Republic of the Philippines

North Eastern Mindanao State University

CogniSync: Addressing Philippine Student Tech Addiction Through Web-Based Generative AI Intervention

IT-related issue: Tech Addiction and Mental Health

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Project Rationale

The rapid advancement and integration of digital technologies have fundamentally transformed the educational and social landscapes of the 21st century. In the Philippines, this digital shift is particularly pronounced, as Filipino youth are consistently ranked among the heaviest users of the internet and social media globally (Kemp, 2024). While internet connectivity provides students with unprecedented access to learning materials and facilitates vital communication, its pervasive availability has led to an alarming rise in excessive and compulsive usage. The increased reliance on digital platforms, accelerated by the recent shift to hybrid learning environments (Cleofas & Rocha, 2021), has blurred the lines between productive academic use and recreational overuse, elevating tech addiction to a major public health concern for Filipino students (Abad Santos & Sarile, 2023).

Internet and smartphone addiction are increasingly recognized within clinical literature as non-substance behavioral addictions characterized by an uncontrollable, impulsive need to connect to digital networks (Montag et al., 2021). For college students, smartphones serve a dual function as both essential academic tools and primary sources of entertainment, making boundary-setting incredibly difficult. Local studies emphasize that factors such as daily screen time and social environment significantly predict smartphone addiction prevalence among Filipino youth (Buctot et al., 2020). Recent surveys further corroborate this, indicating that Filipino university students are heavily reliant on their devices, often engaging with screens immediately upon waking and right before sleeping (Giray et al., 2024). This lack of behavioral self-control manifests as a compulsive need to stay connected, progressively leading adolescents and young adults to prioritize virtual environments over face-to-face social engagements and critical academic activities.

The psychological ramifications of this dependency are profound, with a growing body of literature establishing a direct link between tech addiction and deteriorating mental health. Extensive engagement with digital devices, particularly social media platforms, exposes students to significant "digital stress." This stress often stems from approval anxiety, connection overload, and a pervasive fear of missing out (FOMO) on virtual social validation a phenomenon globally recognized as a primary mediator between problematic smartphone use and anxiety (Elhai et al., 2020; Giray et al., 2024). As students increasingly rely on virtual interactions for emotional fulfillment, the displacement of genuine, in-person social support networks frequently leads to profound feelings of loneliness, chronic psychological stress, and depressive symptoms.

Beyond general psychological distress, smartphone dependency specifically impairs the self-perception and emotional well-being of the youth. In the digital space, social media algorithms continuously expose users to curated and unrealistic portrayals of beauty and lifestyle. In the Philippines, the negative psychological effects of this exposure are magnified; Filipino adolescents who are deeply embedded in social networking sites are highly vulnerable to negative self-perceptions, decreased self-esteem, body image distortion, and appearance-related anxiety when subjected to an endless stream of idealized content (Lacsa, 2025). This reliance creates a vicious cycle wherein students utilize digital platforms to cope with their distress, thereby exacerbating their addictive behaviors and further degrading their emotional resilience.

The compounding effects of digital stress, psychological exhaustion, and diminished self-esteem inevitably spill over into the academic domain, severely compromising scholastic performance. The constant distractions afforded by digital devices deplete the cognitive resources required for sustained attention and complex decision-making, leading to poor time management and diminished academic engagement. When students are chronically exhausted and emotionally drained by their digital habits often worsened by severe sleep deprivation caused by late-night screen time their overall academic self-efficacy plummets (Giray et al., 2024). This cognitive depletion directly increases the likelihood of academic neglect and failure.

Given the pervasive nature of tech addiction and its multi-faceted threat to student well-being in the country, there is an urgent need to establish comprehensive interventions and policies within Philippine educational institutions. The undeniable link between digital dependence, psychological distress, and academic decline highlights a critical gap in current student support systems. Addressing this issue requires a multifaceted approach that not only limits excessive digital exposure but also promotes digital and media literacy, healthy online habits, and robust psychological support through structured digital detox and cognitive interventions (Radtke et al., 2022). Developing localized technical solutions and institutional policies tailored to mitigate tech addiction is essential for safeguarding the holistic well-being and academic success of the modern Filipino student population (Lacsa, 2025).

CogniSync addresses the mental health challenges specifically driven by tech addiction by providing a specialized web-based Generative AI intervention tool for Filipino high school and college students. Operating as an interactive digital wellness coach, the system helps students manage their digital habits when they feel a strong urge to scroll. To ensure interventions are evidence-based and culturally relevant, CogniSync utilizes Retrieval-Augmented Generation (RAG) backed by a curated database of Philippine mental health research. The system is entirely non-personalized; it does not track, store, or adapt to individual user behavior, maintaining a standardized intervention approach that protects student privacy and anonymity. Rather than acting as a simple content filter that blocks apps or providing medical diagnoses, CogniSync equips students with the mental tools for self-regulation. It achieves this through standardized cognitive strategies like CBT-Lite prompts and exit strategies that suggest popular Philippine hobbies like local sports and board games. Ultimately, by enforcing a strict domain limitation focused solely on mental digital addiction, CogniSync aims to safely help students build resilience and cognitive discipline in a world full of digital distractions.

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